Second Assessment

Mandip Chaudhary (17)

Load necessary libraries

```
library(ggplot2)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

6. Do the following using R studio using ggplot2 and dplyr packages with R script to knit PDF file

a. Create a dataset with following variables: age (25-59 years), height (110-190 centimeters), weight (40-90 kg) with random 150 cases of each variable.

```
set.seed(17)  # Roll number
age <- sample(25:59, 150, replace = TRUE)
height <- sample(110:190, 150, replace = TRUE)
weight <- sample(40:90, 150, replace = TRUE)

bmi_data <- data.frame(age, height, weight)
```

b. Compute body mass index (BMI) variable as: BMI = [(weight in kg) / (height in m)^2]

```
bmi <- weight / ((height / 100) ^ 2)

bmi_data <- bmi_data %>%
  mutate(bmi = bmi)
```

c. Create a body mass categories: <18, 18-24, 25-30, 30+ and label them as "Underweight", "Normal", "Overweight", "Obese" respectively using dplyr package

```
bmi_category <- case_when(
  bmi < 18 ~ "Underweight",
  bmi >= 18 & bmi < 25 ~ "Normal",
  bmi >= 25 & bmi < 30 ~ "Overweight",
  TRUE ~ "Obese"
)

bmi_data <- bmi_data %>%
  mutate(bmi_category = bmi_category)
```

d. Show the percentage distribution of labelled BMI variable with pie chart using ggplot2 package

```
pie_data <- bmi_data %>%
  group_by(bmi_category) %>%
  summarise(count = n()) %>%
  mutate(percentage = count / sum(count) * 100)

ggplot(pie_data, aes(x = "", y = percentage, fill = bmi_category)) +
  geom_bar(stat = "identity", width = 1) +
  coord_polar("y") +
  labs(title = "Percentage Distribution of BMI Categories") +
  theme_void() +
  theme(legend.title = element_blank())
```